

RedKross Unified Threat Intelligence Platform Project

Requirements & Documentation

1.Executive summary

Remcos (Remote Control and Surveillance) is a commercially available Remote Access Trojan (RAT) first observed in the wild in 2016. Initially marketed by Breaking Security as a legitimate Windows remote management utility, it was rapidly weaponised by cybercriminals and nation-state-aligned threat actors. Remcos grants full control over infected Windows systems, enabling keylogging, screen and webcam capture, audio recording, file manipulation, credential theft, data exfiltration, and remote code execution. As of 2025, Remcos doubled its activity year-over-year and ranked among the top-four most detected malware families globally with approximately 16,000 detections. It is actively deployed by multiple threat actors, including APT-C-36 (Blind Eagle), UAC-0050, and various financially motivated_cybercriminal clusters. The malware continues to evolve rapidly, incorporating fileless execution, steganographic concealment, and real-time live C2 surveillance capabilities

2.Scope of the Platform

The platform is designed to provide centralized threat intelligence and malware analysis capabilities focused on Remcos RAT and related cyber threats. It supports tracking malware activity, campaign analysis, IOC correlation, and threat actor profiling in a scalable and searchable environment.

Platform Scope

- Remcos RAT lifecycle intelligence and historical campaign tracking
- IOC management including:
 - IP addresses
 - Domains
 - URLs
 - File hashes
 - DNS records
 - Certificates

- Mutexes

Threat actor profiling:

- Aliases
- Motivation
- Infrastructure
- Tactics, Techniques, and Procedures (TTPs)

Campaign associations

Malware family intelligence for Remcos RAT:

- Discovery history
- Technical behavior
- Persistence methods
- Exfiltration techniques
- YARA rules

Campaign tracking and timeline analysis

- MITRE ATT&CK mapping for Remcos RAT activities
- Command and Control (C2) infrastructure analysis
- Victim country and target industry analysis

Detection engineering support:

- YARA rules
- Sigma rules
- IOC enrichment
- Threat activity monitoring for newly observed campaigns
- Malware sample and artifact analysis
- Reporting and intelligence documentation
- Evidence management and research note storage
- Integration with threat intelligence feeds and malware sandboxes

Main Objectives

- Improve malware investigation efficiency
- Correlate Remcos RAT campaigns with threat actors
- Detect malicious infrastructure and phishing operations
- Support incident response and threat hunting
- Maintain a searchable intelligence repository

- Provide actionable cybersecurity intelligence for defenders

Expected Outputs

- Malware family profile pages
- IOC knowledge base
- Threat actor intelligence reports
- Campaign timelines
- Detection rule libraries
- Threat severity assessments
- Infrastructure analysis reports
- Research summaries and evidence tables

3.Core Intelligence Modules

	Module	Purpose	Key Outputs
	APT Intelligence Engine	Tracks threat actors, campaigns, and attribution related to Remcos RAT.	APT profile, campaign timeline, related groups
	IOC Correlation Engine	Links hashes, IPs, domains, DNS records, URLs, and mutexes.	IOC graph, enrichment results, linked infrastructure
	Malware Family Module	Stores Remcos RAT history, behavior, TTPs, and malware analysis.	Malware family page, YARA rules, malware notes
	Threat Activity Monitor	Detects new campaigns and infrastructure changes.	Alerts, campaign updates, activity monitoring
	Campaign Tracking Module	Tracks malware campaigns, victim countries, and industries.	Campaign reports, attack timeline, victim analysis
	MITRE ATT&CK Mapping Module	Maps Remcos RAT behavior to ATT&CK techniques.	ATT&CK mappings, adversary behavior analysis
	Infrastructure Analysis Module	Analyzes C2 servers, DNS, domains, hosting, and IPs.	Infrastructure reports, DNS analysis, IP tracking
	Detection Engineering Module	Develops detection logic for Remcos RAT activity.	YARA rules, Sigma rules, detection signatures
	Threat Intelligence Dashboard	Provides centralized and searchable intelligence.	Dashboards, IOC summaries, statistics
	Research & Evidence Workspace	Stores malware reports, screenshots, and analyst notes.	Research notes, evidence tables, malware documentation

The Core Intelligence Modules are the main components of a cyber threat intelligence system used for analyzing and tracking Remcos RAT malware activities.

- APT Intelligence Engine – Tracks threat actors and campaigns.
- IOC Correlation Engine – Links hashes, IPs, domains, and URLs.
- Malware Family Module – Stores malware behavior and analysis data.
- Threat Activity Monitor – Detects new malware activities and changes.
- Campaign Tracking Module – Tracks campaigns, victims, and industries.
- MITRE ATT&CK Mapping Module – Maps malware behavior to ATT&CK techniques.
- Infrastructure Analysis Module – Analyzes C2 servers, DNS, and IPs.
- Detection Engineering Module – Creates YARA, Sigma, and detection rules.
- Threat Intelligence Dashboard – Displays centralized threat data and statistics.
- Research & Evidence Workspace – Stores reports, screenshots, and analyst notes.

4.ATP Intelligence data model

APT Intelligence Data Model		
Field		Description
	APT Name and Aliases	Official threat actor name and known aliases associated with Remcos RAT campaigns.
	Life-Time / Active Period	Duration of activity from first observed to latest observed operations.
	Threat Actor Type and Motivation	Classification such as cybercriminal, espionage, financially motivated, or hacktivist.
	Known Operators / Public References	Information about known individuals, groups, or public intelligence references.
	MITRE ATT&CK TTPs	Techniques and tactics used during attacks mapped to MITRE ATT&CK.
	IOC List	IPs, domains, hashes, URLs, mutexes, certificates, DNS records, and user agents.
	Campaign List	List of campaigns with first-seen and last-seen dates.
	Victim Countries	Countries targeted during Remcos RAT operations.
	Target Industries	Industries affected such as banking, healthcare, government, and education.
	Observed Malware Families and Tools	Associated malware, loaders, droppers, and attack tools.
	Infrastructure Notes	Hosting providers, DNS behavior, C2 servers, ASN details, and redirect activity.
	New Activity Timeline	Timeline of newly observed campaigns, payload changes, and infrastructure updates.
	Related APT Groups and Overlaps	Connections or overlaps with other threat actors or campaigns.

The image represents an APT Intelligence Data Model used in Cyber Threat Intelligence (CTI).

It organizes important information related to APT groups and cyber attacks.

The model includes attacker details, malware, campaigns, and victim information.

MITRE ATT&CK mapping helps analyze attacker tactics and techniques.

IOC data such as IPs, domains, and hashes supports threat detection.

Infrastructure and timeline analysis help track evolving cyber threats.

The model improves threat hunting, monitoring, and incident response activities.

5.Campaign Tracking Fields

CAMPAIGN TRACKING								
Campaign ID	Threat Actor	Activity Period (UTC)	Victim Country	Target Industry	TTPs	IOC	Summary	Notes
								
CAMP-001	Remcos RAT Operators	Jan 2024 ~ Mar 2024	India	Banking	T1566, T1059	IPs, domains, hashes	Phishing campaign delivering Remcos RAT through invoice attachments	Financial credential theft observed
CAMP-002	Unknown Threat Group	Apr 2024 ~ Jun 2024	United States	Healthcare	T1204, T1547	URLs, hashes, mutexes	Malicious Office documents used for malware delivery	Persistence established through registry keys
CAMP-003	Cybercriminal Group	Jul 2024 ~ Sep 2024	Germany	Government	T1055, T1105	Domains, IPs	Remote access and surveillance activity detected	Encrypted C2 communication observed
CAMP-004	Financially Motivated Actors	Oct 2024 ~ Dec 2024	United Kingdom	Education	T1027, T1005	Hashes, URLs	Obscured Remcos payload distributed via phishing emails	Data exfiltration and keylogging activity

- * The framework standardizes the collection of campaign-related threat intelligence data.
- * It helps analysts document and monitor cyber attack campaigns efficiently.
- * Campaign IDs provide unique identification for tracking threat activities.
- * Threat actor information helps associate campaigns with known attacker groups.
- * Activity period fields help identify attack duration and timelines.
- * Victim country and target industry fields show affected regions and sectors.
- * TTPs map attacker behavior using the MITRE ATT&CK framework.
- * IOC fields store malicious IPs, domains, hashes, and URLs for detection.
- * Summary and notes provide a quick overview of campaign activities.
- * The model supports better threat hunting, reporting, and incident response operations.

6.Malware Family Documentation

Family Name and Aliases

- Remcos RAT
- Remcos
- Remcos Remote Access Trojan

Discovery History and Timeline

- First discovered around 2016
- Originally sold as a legitimate remote administration tool
- Later weaponized and used in malicious phishing campaigns
- Continues to be actively used in cybercrime operations worldwide

Original Purpose and Evolution

- Designed for remote system administration
- Later modified by attackers for:
 - Espionage
 - Credential theft
 - Surveillance
 - Remote control
 - Data exfiltration

Technical Characteristics and Behavior

- Remote desktop access
- Keylogging
- Clipboard monitoring
- Screenshot capture
- Webcam and microphone access
- Command execution
- File management
- Process injection
- Anti-analysis and obfuscation techniques

Common File Extensions and Packaging Formats

File Extensions

- .exe
- .dll
- .scr
- .bat
- .vbs

Packaging Formats

- ZIP archives
- RAR archives
- ISO files
- Office macro documents
- MSI installers

Known Hashes, Signatures, and YARA Rules

- Sample Hashes
- SHA256 hashes
- MD5 hashes
- File signatures

YARA Detection

- Detects Remcos-specific strings
- Identifies mutex patterns
- Detects persistence artifacts

Associated APT Groups and Campaigns

- Financial cybercriminal groups
- Phishing operators
- Fake invoice campaigns
- Malware delivery operations
- Malspam campaigns

Typical TTPs and Infection Chain

MITRE ATT&CK Techniques

- T1566 – Phishing
- T1059 – Command and Scripting Interpreter
- T1547 – Persistence
- T1055 – Process Injection
- T1105 – Ingress Tool Transfer

Infection Flow

- 1. Phishing email delivery
- 2. Malicious attachment opened
- 3. Script or macro execution
- 4. Payload download
- 5. Persistence establishment
- 6. C2 communication
- 7. Credential theft and surveillance

Network Indicators, Domains, and IPs

- Malicious IP addresses
- Dynamic DNS domains
- Command-and-Control servers
- Encrypted HTTPS traffic
- VPS-hosted infrastructure

Persistence, Privilege Escalation, and Exfiltration Behavior

Persistence

- Registry Run keys
- Startup folders
- Scheduled tasks

Privilege Escalation

- UAC bypass attempts

- Token manipulation

Exfiltration

- Browser credentials
- Clipboard data
- Screenshots
- Documents and files

Unique File System or Artifact Patterns

- Mutex creation
- Registry modifications
- Temporary payload files
- Scheduled task artifacts
- AppData persistence files

Command and Control (C2) Infrastructure

- Dynamic DNS providers
- Encrypted communication channels
- Remote VPS infrastructure
- HTTPS-based C2 traffic

Threat Actor Association

- Financially motivated cybercriminal groups
- Commodity malware operators
- Phishing-based threat actors

Attribution Confidence

Medium to High confidence based on:

- Malware behavior
- Infrastructure overlap
- Threat intelligence reporting

- MITRE ATT&CK technique mapping

Indicators of Compromise (IOCs)

- Domains
- IP addresses
- File hashes
- Registry keys
- Mutex names
- User-agent strings

Threat Severity

High Severity

Reasons

- Full remote access capability
- Credential theft
- Persistent access
- Surveillance functionality
- Easy phishing deployment

Victim Countries

- India
- United States
- Germany
- United Kingdom

Target Industries

- Banking
- Healthcare
- Government
- Education
- Manufacturing

- Information Technology

Detection Opportunities

- Monitor PowerShell execution
- Detect registry persistence changes
- Analyze outbound encrypted traffic
- Detect macro-enabled Office documents
- Monitor scheduled task creation

Defensive Recommendations

- Disable Office macros
- Use Endpoint Detection and Response (EDR)
- Apply email filtering
- Enable Multi-Factor Authentication (MFA)
- Monitor suspicious outbound connections
- Deploy YARA and Sigma rules

Real-World Incidents

- Fake invoice phishing campaigns
- Malicious Office document attacks
- Financial credential theft operations
- Remote surveillance campaigns

7. Ransomware Family Pages

Note: Remcos RAT is primarily a **Remote Access Trojan (RAT)** and not a ransomware family. However, it is often used alongside ransomware operations for initial access, surveillance, persistence, and credential theft before ransomware deployment.

Ransomware Name and Aliases

- Remcos RAT (associated malware tool)
- Remcos
- Remote Access Trojan

First Observed and Latest Observed Activity

- First Observed: Around 2016
- Latest Observed Activity: Ongoing active campaigns observed globally

Common File Extensions Appended to Encrypted Files

Remcos RAT itself does not encrypt files like ransomware.

However, when used with ransomware campaigns, related ransomware families may append:

- .locked
- .encrypted
- .lockbit
- .ryuk
- .conti

Encryption and Extortion Style

Remcos RAT generally supports ransomware operators by:

- Providing remote access
- Delivering ransomware payloads
- Stealing credentials
- Monitoring victims before encryption

Extortion Style

- Double extortion operations
- Data theft before encryption
- Threatening data leaks
- Financial ransom demands

Known Leak Portals / Onion References

Remcos RAT itself has no official leak portal.

However, ransomware groups using Remcos may operate:

- TOR-based leak sites
- Onion negotiation portals
- Victim shaming websites

Associated Threat Actor or Affiliate Model

- Financially motivated cybercriminal groups
- Ransomware affiliates
- Phishing-based malware operators
- Initial Access Brokers (IABs)

Observed TTPs and Target Sectors

Observed MITRE ATT&CK Techniques

- T1566 – Phishing
- T1059 – Command and Scripting Interpreter
- T1547 – Boot or Logon Autostart Execution
- T1055 – Process Injection
- T1027 – Obfuscated Files or Information
- T1105 – Ingress Tool Transfer

Target Sectors

- Banking
- Government
- Healthcare
- Education
- Manufacturing
- Information Technology

Related Samples, Hashes, and Infrastructure

Related Infrastructure

- Dynamic DNS domains
- VPS-based command-and-control servers
- HTTPS-encrypted communication
- Fast-changing malicious IP infrastructure

Sample IOC Examples

Domains

- update-client.ddns.net

- `secure-admin.duckdns.org`

Mutexes

- `RemcosMutex`

Registry Keys

- `HKCU\Software\Remcos`

File Names

- `remcos.exe`
- `svchost.exe`
- `client32.exe`

Victim Countries

- India
- United States
- Germany
- United Kingdom

Threat Severity

High Severity

Reasons

- Enables full remote access
- Frequently used before ransomware deployment
- Supports credential theft and lateral movement
- Helps attackers maintain persistence before encryption stage

8. Library Folder Review Workflow

This example shows how to apply the workflow specifically for Remcos RAT intelligence analysis.

1. Folder Assignment

- **Folder Name:** Remcos_RAT_Intelligence_Folder
- **Owner:** Threat Intelligence Team
- **Review Date:** 19 May 2026
- **Scope:** Malware samples, phishing files, IOC datasets, reports, screenshots

2. File Distribution (100 Files → 10 Students)

- Student 1 → Files 1–10
- Student 2 → Files 11–20
- Student 3 → Files 21–30
- Student 4 → Files 31–40
- Student 5 → Files 41–50
- Student 6 → Files 51–60
- Student 7 → Files 61–70
- Student 8 → Files 71–80
- Student 9 → Files 81–90
- Student 10 → Files 91–100

3. Extract File Details

Each student extracts:

- Filename
- File Type (EXE, DOCM, ZIP, etc.)
- Hash (MD5 / SHA256)
- Date (Created / Modified)
- Source (Email, Sandbox, Malware DB)
- Relevance (High / Medium / Low)

4. File Tagging

Each file is tagged as:

- **Important** → Active malware or critical IOC
- **Suspicious** → Potential malicious behavior
- **Duplicate** → Repeated files or hashes
- **Reference-only** → Reports, articles, screenshots

5. Short Notes

- *remcos_payload.exe* → Active Remcos RAT sample with keylogging capability
- *invoice_macro.docm* → Phishing document delivering malware via macro
- *c2_list.txt* → Contains command-and-control domains
- *registry_dump.txt* → Shows persistence via Run keys
- *screen_capture.png* → Evidence of remote access session

6. Folder-Level Findings

Common patterns observed:

- Heavy use of phishing emails with macro-enabled documents
- Repeated use of dynamic DNS domains for C2
- Persistence via registry Run keys and startup folders
- Keylogging and screenshot capture across multiple samples
- Encrypted communication with attacker servers

7. File Issues Identified

- 5 duplicate malware samples (same hash)
- 3 corrupted ZIP files
- 2 missing payload files referenced in reports
- Repeated IOC lists across multiple documents

8. Final Folder Report

Filename	Type	Hash	Relevance	Tag
remcos_sample.exe	EXE	SHA256 Hash	High	Important
phishing_invoice.docm	DOCM	SHA256 Hash	High	Suspicious
c2_domains.txt	TXT	SHA256 Hash	Medium	Important

screenshot1.png	PNG	N/A	Low	Reference-only
duplicate_sample.exe	EXE	Same Hash	Low	Duplicate

Final Output Summary

The folder review provides:

- Consolidated intelligence on Remcos RAT
- Cleaned and deduplicated dataset
- Identified infection patterns and infrastructure
- Structured evidence for analysis and reporting
- Actionable insights for detection and response

9. Final Deliverables

The final output of the Remcos RAT Threat Intelligence Platform should include the following structured deliverables:

APT Profile Pages

Detailed intelligence profiles of threat actors associated with Remcos RAT, including:

- Aliases and attribution
- Motivation and threat type
- Campaign history
- Infrastructure usage
- MITRE ATT&CK mapping
- Associated malware and tools

Campaign Timeline Dashboard

A visual and analytical representation of:

- Campaign start and end dates
- Activity patterns over time
- Victim countries and industries
- Attack progression and phases

IOC Knowledge Base

A centralized repository of Indicators of Compromise:

- IP addresses
- Domains and URLs
- File hashes (MD5, SHA256)
- Mutexes and registry keys
- DNS and certificate data

Malware Family Pages

Comprehensive documentation of the malware family:

- History and evolution
- Technical behavior and capabilities
- Infection chain and persistence
- Command-and-control communication
- Detection opportunities

Ransomware Family Pages

Dedicated pages for ransomware families linked with Remcos RAT operations:

- Ransomware profile and aliases
- Encryption behavior and file extensions
- Leak sites and onion portals
- Extortion techniques
- Target sectors and victim countries

YARA and Detection Library

A collection of detection rules for identifying Remcos RAT activity:

- YARA rules for malware detection
- Sigma rules for SIEM monitoring
- Behavioral detection patterns
- IOC-based detection logic

Research Notes and Evidence Tables

Structured documentation of analysis findings:

- Malware analysis notes
- Sandbox results
- Screenshots and artifacts
- Evidence tables with file metadata
- Analyst observations

Library-Folder Analysis Reports

Final summarized reports for each reviewed folder:

- File-level analysis and tagging
- Pattern identification
- Duplicate and missing file detection
- Intelligence insights and conclusions
- Evidence tables and summaries

Final Outcome

These deliverables collectively provide:

- A complete intelligence ecosystem for Remcos RAT
- Actionable insights for cybersecurity teams
- Support for threat hunting, incident response, and detection engineering
- A structured and searchable intelligence repository